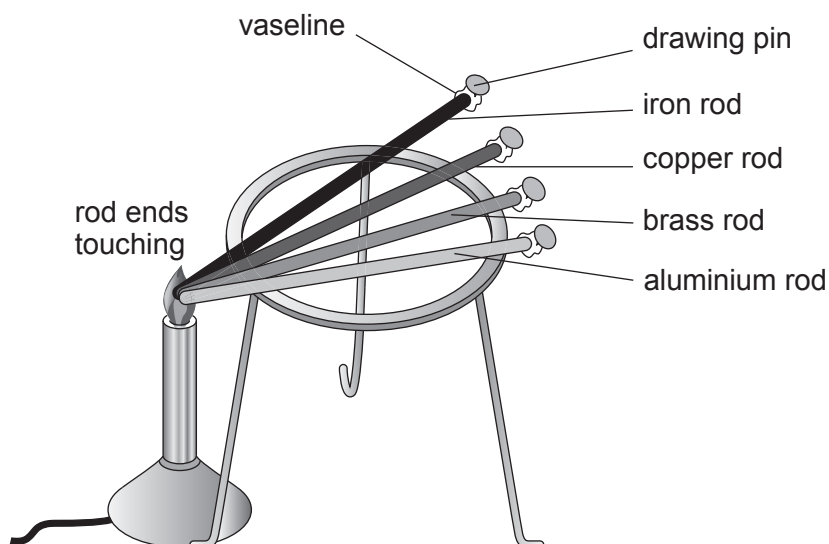


3. Metals are good conductors of heat and electricity.

(a) Three students investigated the thermal conductivity of four metal rods as shown below.



State **two** ways in which the students will make the experimental method valid.

[2]

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(b) The table shows results from the students' investigation together with theoretical values of thermal conductivity.

Metal	Time taken for pin to drop (s)	Thermal conductivity (W/m K)
iron	17	80.4
copper	7	401
brass	12	109
aluminium	5	210

Use the data to explain whether the students' results are as expected.

[3]

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- (c) Explain, in terms of structure and bonding, why the electrical conductivity of sodium chloride is dependent upon its state. [3]

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4. Food self-sufficiency means that a country produces enough food to feed all of its people without having to import food from other countries.

- (a) Discuss the impact of using fertilisers **and** pesticides in farming. [4]

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- (b) Students investigated the effect of one type of NPK fertiliser.

Rice plants were treated with solutions of different concentrations of this NPK fertiliser. They were treated each day for five weeks.

The growth of each plant was recorded and the results are shown in the table below.

Concentrations of NPK fertiliser (units)	Plant growth after 5 weeks (mm)				
	plant 1	plant 2	plant 3	plant 4	mean
0.2	27	29	32	28	29
0.4	55	57	58	62	58
0.6	65	67	68	72	68
0.8	70	72	73	77	73



- (i) Calculate the difference in mean plant growth **per week** for the plants treated with 0.2 and 0.6 units of NPK fertiliser. [3]

difference = mm/week

- (ii) John suggests that mean growth **always** doubles if the concentration of fertiliser is doubled. Use data from the table to explain whether you agree with him. [3]

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- (c) Describe how plant breeding programmes are used to produce crops with higher rice yields. [4]

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- (d) Hydroponics is used to grow plants under controlled conditions. Describe how hydroponics differs from traditional methods of growing plants. [2]

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